

Assignment No 11

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Question:

Write a program to create a Book class and extend it by using dynamic memory allocation. Use the new keyword and object pointers to create Book objects dynamically. The program should allow the user to: Input details of one or more books. Display the book details. Demonstrate the use of dynamic memory allocation for objects.

Code:

```
/*
Write a program to create a Book class and extend it by using dynamic memory allocation.
Use the new keyword and object pointers to create Book objects dynamically. The program
should allow the user to: Input details of one or more books. Display the book details.
Demonstrate the use of dynamic memory allocation for objects..
*/

#include<iostream>
#include<limits>
using namespace std;

class Book{
private:
    string title;
    string author;
    float price;

public:
    void inputDetails(){
        cout << "Enter Book Title: ";
        cin.ignore();
        getline(cin, title);

        cout << "Enter Author Name: ";
        getline(cin, author);

        while(true){
            cout << "Enter Price: ";
            cin >> price;

            if(cin.fail()){
                cin.clear();
                cin.ignore(numeric_limits<streamsize>::max(), '\n');
                cout << "Invalid input. Enter numeric value.\n";
            }
            else if(price < 0){
                cout << "Price cannot be negative.\n";
            }
            else{
                break;
            }
        }
    }
}
```

```

        void displayDetails(){
            cout << "\nBook Details\n";
            cout << "Title   : " << title << endl;
            cout << "Author  : " << author << endl;
            cout << "Price   : " << price << endl;
        }
};

int main(){
    int n;

    while(true){
        cout << "Enter number of books: ";
        cin >> n;

        if(cin.fail() || n <= 0){
            cin.clear();
            cin.ignore(numeric_limits<streamsize>::max(), '\n');
            cout << "Invalid input. Enter positive integer.\n";
        }
        else{
            break;
        }
    }

    Book* books = new Book[n];

    for(int i = 0; i < n; i++){
        cout << "\nEnter details for Book " << i + 1 << endl;
        books[i].inputDetails();
    }

    cout << "\n--- Book Records ---\n";

    for(int i = 0; i < n; i++){
        books[i].displayDetails();
    }

    delete[] books;

    return 0;
}

```

Output:

```
PS D:\STUDY\B.Tech\SEM IV\OOP\Assignment 11> .\a.exe
Enter number of books: 2

Enter details for Book 1
Enter Book Title: words of wisdom
Enter Author Name: om varma
Enter Price: 90

Enter details for Book 2
Enter Book Title: wisdom of words
Enter Author Name: varma om
Enter Price: 09

--- Book Records ---

Book Details
Title  : words of wisdom
Author : om varma
Price  : 90

Book Details
Title  : wisdom of words
Author : varma om
Price  : 9
PS D:\STUDY\B.Tech\SEM IV\OOP\Assignment 11> █
```